MATH INQUIRERS PLACEMENT CHECK

Ages 11-13

For children after the PSLE. Open to a year either side • About 30 minutes • No calculator

Name or username: _____ Age: _____ Date: _____

How it works. This is a friendly check, not a test. Your child answers on their own. Read a question aloud if they ask, but do not help with the maths. If they are stuck, say "Try your best, or skip it". Working on scrap paper is fine.
Not learnt it yet? Some questions may use ideas your child has not met at school. That is fine and useful to know. Tick "not learnt yet" and move on.

Warm-up

1 Write every factor of 24 (every whole number that divides 24 exactly).

☐ not learnt yet

2 Write 6/36 in its simplest form.

☐ not learnt yet

3 Two angles of a triangle are 52 and 63 degrees. What is the third angle, in degrees?

☐ not learnt yet

4 Read this little program. Start with n = 5. Then repeat 4 times: change n by 3 (add 3 to n). At the end, say n. What does it say?

☐ not learnt yet

Main questions

5 A number machine doubles the input and then adds 5. The output is 31. What was the input?

not learnt yet

6 A prime number has exactly two factors: 1 and itself. For example, 7 is prime, but 9 is not (its factors are 1, 3 and 9). How many prime numbers are there between 30 and 50?

not learnt yet

7 The temperature is -4 degrees C at night. It rises by 9 degrees by noon, then falls by 12 degrees by evening. What is the evening temperature in degrees C?

not learnt yet

8 A bag has 4 red, 3 blue and 5 green marbles. One marble is picked without looking. The chance (probability) of picking blue is 3 out of 12, written 3/12. What is the probability that it is NOT green? Give a fraction in simplest form.

not learnt yet

9 The average of 5 numbers is 12. Four of the numbers are 8, 10, 15 and 11. What is the fifth number?

not learnt yet

10 It is 10 o'clock now on a 12-hour clock. What time will it be 17 hours later? Give the hour only.

not learnt yet

Brain stretchers

Optional. These are meant to be tricky, and most children will not finish them. Have a go at any that look fun.

11 The pattern 4, 7, 10, 13, ... carries on in the same way. What is the 50th number? Explain how you worked it out.

not learnt yet

12 Using the digits 1, 2, 3 and 4, with no digit used more than once, how many different 3-digit even numbers can be made?

not learnt yet

YOUR CHILD'S RESULT

Which level fits?

Mark the answers with the key (ask us for it, or use the tutor key). Add up the marks, then find your row. This is a guide. You make the final choice, and you can change level free of charge before lesson 1.

Warm-up: ____ / 4 **Main questions:** ____ / 6 **Brain stretchers (optional):** ____ / 2
Ticked "not learnt yet": ____

Main questions	What it suggests
4 to 6	Join **Ages 11-13 (Level 6)**. Your child has the foundations and will enjoy the thinking.
2 or 3	A short chat with a tutor and the free taster lesson will show which level feels best. Ages 11-13 (Level 6) with a little support suits many children here. If the warm-up score is 2 or less, Ages 10-12 (Level 5) may be the more comfortable start.
0 or 1	We suggest **Ages 10-12 (Level 5)**. Ideas build step by step, and a confident start matters more than a fast one. If many questions were ticked "not learnt yet", that simply means the topics have not been taught yet.

Finding it easy?
Choose the level that fits the score, not the school year. Every level has open investigations for children who want more.

Found it hard?
That is useful to know, and it says nothing about how clever your child is. A level that feels comfortable builds confidence, and your child can always move up.

Skills covered

Number patterns and rules; Factors and primes; Negative numbers; Algebra machines; Counting possibilities; Probability; Turns and coding with loops; Clock arithmetic and codes; Average and data

Next step. Send us the scores and your preferred level when you enrol. Prices are the same at every level, and a tutor can talk it through first. Small groups of up to 8, live online.